



Technical Research Report: The Modern Computer Vision Vision Pipeline

Prepared by: Muhammad Abdullah Nazir

Program: BS Artificial Intelligence

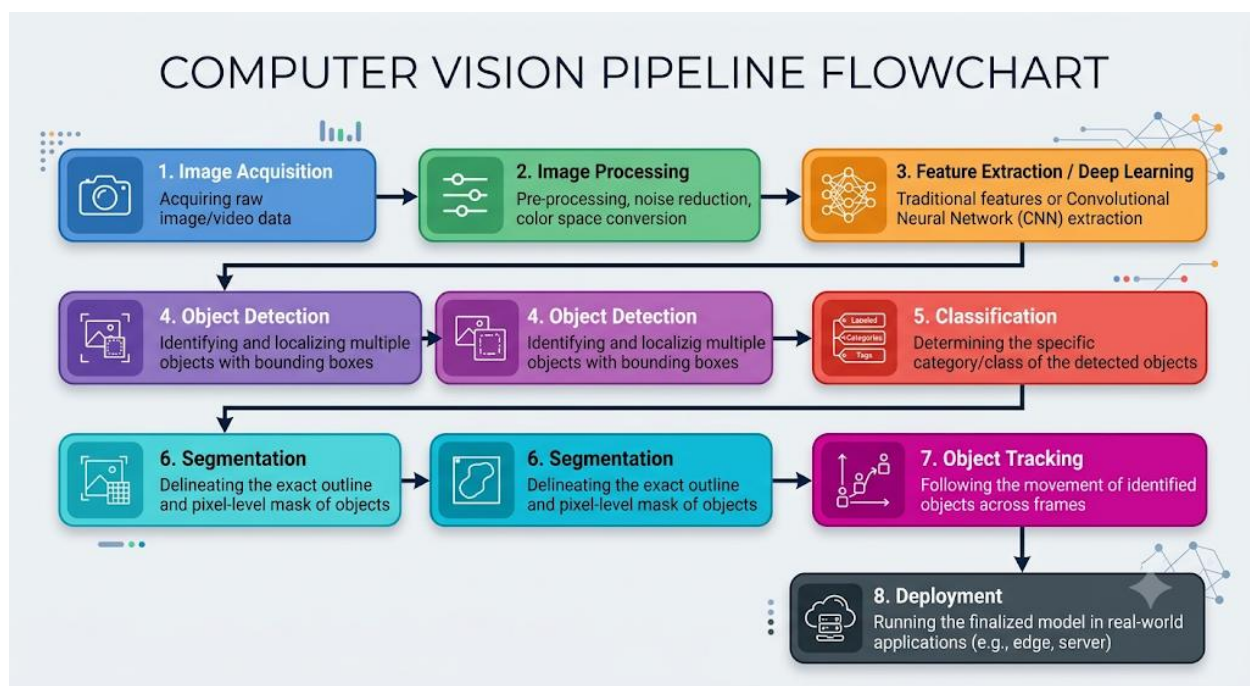
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Abstract

Computer Vision (CV) is a branch of Artificial Intelligence (AI) that enables computers to interpret and understand visual information from images and videos. Modern computer vision systems combine image processing, deep learning, and advanced object detection techniques to automate tasks that previously required human vision. Applications such as autonomous vehicles, medical image diagnosis, facial recognition, industrial automation, agriculture, and surveillance systems rely heavily on computer vision technologies

1. Modern Computer Vision Pipeline

The modern computer vision workflow consists of several interconnected stages.



Each stage contributes to improving the accuracy and efficiency of AI-powered vision systems.

2. Image Acquisition

Image acquisition is the first step in the computer vision pipeline. It involves collecting images or video data from various sources that will later be processed and analyzed.

Sources of Image Acquisition

- Digital Cameras
- CCTV Cameras
- Mobile Phones

- Drones
- Medical Imaging Devices (X-ray, MRI, CT Scan)
- Satellite Sensors
- Industrial Cameras

The quality of collected data directly affects model performance. Images should have sufficient resolution, proper lighting, and represent different viewing angles and environmental conditions.

3. Image Processing

Image processing prepares raw images before they are used for machine learning or deep learning models.

Common preprocessing techniques include:

- Image resizing
- Noise removal
- Image normalization
- Brightness and contrast adjustment
- Image enhancement
- Data augmentation

These operations improve image quality and allow AI models to learn more effectively.

Common OpenCV Functions

- `cv2.imread()`
- `cv2.resize()`
- `cv2.cvtColor()`
- `cv2.GaussianBlur()`
- `cv2.Canny()`

4. Object Detection

Object detection identifies both the category of an object and its location within an image.

Unlike image classification, object detection predicts multiple objects simultaneously using bounding boxes and confidence scores.

Popular object detection models include:

- YOLO (You Only Look Once)
- Faster R-CNN

Output of Object Detection

For every detected object, the model predicts:

- Object Class
- Bounding Box
- Confidence Score

5. Image Classification

Image classification assigns one or more labels to an entire image.

The model determines the category of the input image without identifying the exact location of objects.

Popular CNN architectures include:

- ResNet
- MobileNet
- VGG
- DenseNet

6. Image Segmentation

Segmentation divides an image into meaningful regions by assigning labels to individual pixels.

Unlike object detection, segmentation provides highly detailed information about object boundaries.

Types of segmentation include:

- Semantic Segmentation
- Instance Segmentation

Popular segmentation models:

- U-Net
- Mask R-CNN
- DeepLabV3+

7. Object Tracking

Object tracking follows detected objects across consecutive video frames while maintaining a unique identity for each object.

Tracking algorithms include:

- DeepSORT
- ByteTrack
- SORT
- Optical Flow

Tracking enables AI systems to monitor object movement over time.

8. Deployment

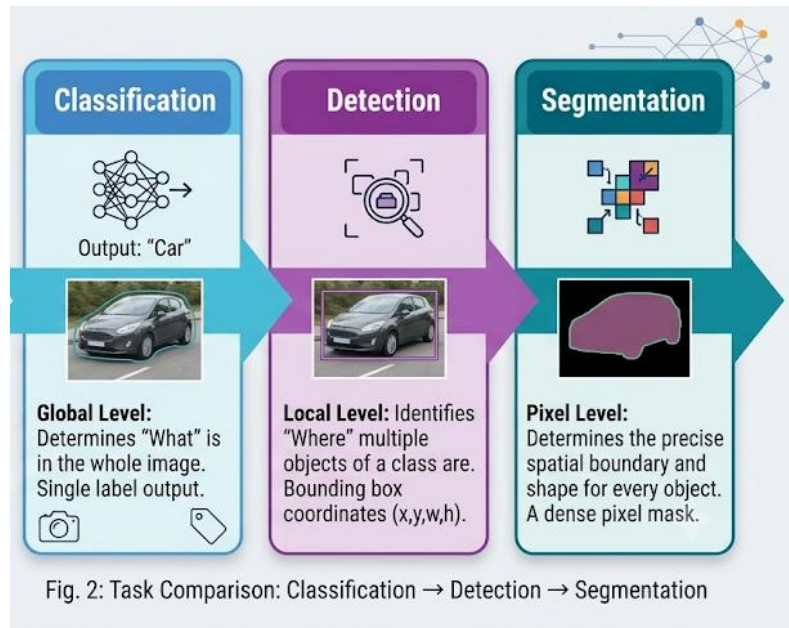
Deployment is the final stage where the trained AI model is integrated into real-world applications.

Deployment platforms include:

- Cloud Servers
- Mobile Applications
- Embedded Systems
- Autonomous Robots
- Smart Cameras

Frameworks commonly used for deployment:

- TensorFlow Lite
- Docker



Real-World Computer Vision Applications

Modern computer vision is transforming numerous industries.

Healthcare

- Disease diagnosis
- Tumor detection
- Medical image analysis

Transportation

- Autonomous vehicles
- Traffic monitoring
- Lane detection

Retail

- Inventory management
- Customer behavior analysis
- Self-checkout systems

Agriculture

- Crop disease detection
- Fruit counting
- Precision farming

Security

- Face recognition
- Smart surveillance

Manufacturing

- Defect detection
- Quality inspection
- Industrial automation

Conclusion

The modern computer vision pipeline is a structured process that converts raw visual data into intelligent decisions. Beginning with image acquisition and progressing through preprocessing, object detection, classification, segmentation, tracking, and deployment, each stage plays a critical role in developing reliable AI systems. With the rapid advancement of deep learning models such as CNNs and YOLO, computer vision continues to revolutionize healthcare, transportation, manufacturing, agriculture, and many other industries. As computing power and AI algorithms continue to improve, computer vision will become even more accurate, efficient, and widely adopted in solving real-world challenges.

References

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